

Muhammad Mostafa Saadawy Amin

Junior Data Engineer

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Professional Summary

Junior Data Engineer and final-year Computers & AI student with hands-on experience building scalable batch and streaming ETL pipelines in Python, Spark, Kafka, and Airflow, following industry standards for reliable pipeline design. Skilled in Big Data analytics, designing Star Schema data warehouses, improving query performance by 30–40%, and deploying analytics solutions on Microsoft Azure. Strong problem-solving skills, analytical thinking, and creativity developed through independent, end-to-end project work, paired with clear communication and interpersonal skills built through collaboration with cross-functional and academic teams. Focused on real-time data processing and CI/CD automation at scale.

Professional Experience

Huawei Big Data Associate Trainee | Huawei ICT Academy

Jul 2026 – Sep 2026

- Built foundational skills in Linux (command line, file systems, permissions), SQL, and Python (data structures, functions) using Jupyter Notebook for data processing.
- Studied HDFS distributed storage, ZooKeeper cluster coordination, and MapReduce/YARN for distributed batch processing and resource management.
- Explored the Hadoop ecosystem including Hive, HBase, and Elasticsearch, alongside Spark's in-memory computing and Flink's stream-batch engine.
- Covered real-time ingestion tools Flume and Kafka within a complete Big Data pipeline architecture.

Microsoft Data Engineer – Trainee (DEPI Scholarship) | MCIT

Jun 2025 – Dec 2025

- Built 10+ batch and streaming ETL pipelines using Python, SQL, Spark, Kafka, and Airflow for data ingestion and transformation.
- Designed 5+ Star Schema data warehouses and optimized analytical queries, improving performance by 30–40%.
- Processed millions of records using Hadoop, Hive, and Spark for scalable Big Data processing.
- Developed cloud data solutions using Azure Synapse, Azure Data Factory, and Azure Stream Analytics, with GitHub Actions for CI/CD automation.

Machine Learning Trainee | National Telecommunication Institute (NTI)

Aug 2025 – Sep 2025

- Trained and evaluated supervised and unsupervised models using Python, NumPy, Pandas, and Scikit-learn.
- Applied data preprocessing, feature engineering, and model tuning to improve accuracy.
- Built end-to-end ML workflows from data collection to model training, evaluation, and deployment via REST APIs using Streamlit.

Projects

Big Data Project – Distributed Big Data & ML Platform

Aug 2026

[GitHub](#) | [Live Demo](#)

- Built an end-to-end distributed Big Data platform for NYC FHVHV trip duration prediction using a 4-device K3s Kubernetes cluster over a private Tailscale mesh network.
- Implemented distributed ETL and ML workflows with Spark/PySpark across 3 worker nodes (32 CPU cores, ~20 GiB RAM).
- Designed a Bronze/Silver/Gold Data Lake using MinIO, orchestrated with Apache Airflow.
- Integrated Kafka, Spark Thrift Server, PostgreSQL, JupyterLab, and Streamlit for streaming, BI connectivity, and visualization.

Technologies: Kubernetes, Apache Spark, PySpark, Apache Airflow, Apache Kafka, MinIO, Python, PostgreSQL, Docker, Tailscale, Streamlit

IKEA Shopping – Admin Panel + Data Engineering & ML Pipeline

Apr 2026 – Jun 2026

[GitHub](#) | [Live Demo](#)

- Extended a Java Swing admin panel (SQL Server, 10-table schema, RBAC, SHA-256 hashing) with a full data engineering and ML layer, without modifying existing Java code.
- Built a synthetic data generator producing 18 months of sales data with seasonality and growth trend using NumPy & PyOD.
- Designed ETL pipeline (SQL Server → Star Schema → SQLite warehouse) with Fact_Sales, Dim_Product, and Dim_Date tables.
- Trained an XGBoost forecasting model with lag features and rolling means; deployed a live Streamlit dashboard for actual vs. forecasted sales.

Technologies: Java, Swing, SQL Server, Python, Pandas, NumPy, XGBoost, Streamlit, SQLite, ETL, Data Warehousing.

Smart City – Real-Time Data Pipeline

Dec 2025 – Apr 2026

[GitHub](#)

- Built a real-time pipeline simulating 5 smart city IoT sources (vehicles, GPS, cameras, weather, emergencies) using Python producers and Kafka.
- Processed concurrent Kafka streams with Spark Structured Streaming, applying schemas, JSON deserialization, and watermarking for late events.
- Containerized the system (Zookeeper, Kafka broker, Spark master + 2 workers) using Docker Compose for reproducible deployment.
- Modeled data entities via ERD across 5 interconnected tables for downstream analytics.

Technologies: Python, Apache Kafka, Spark Structured Streaming, Docker, Docker Compose, Apache Parquet

Road Collisions Severity Prediction – ML Project

Sep 2025 – Nov 2025

[GitHub](#) | [Live Demo](#)

- Built an ML system to predict road collision severity (92.6% accuracy) using real-world traffic data.
- Performed data cleaning, feature engineering, and EDA to identify key severity factors.
- Trained a CatBoost classifier achieving a 0.92 weighted F1-score with perfect precision on Fatal and Serious classes.
- Deployed an interactive Streamlit app for real-time severity predictions.

Technologies: Python, CatBoost, Scikit-learn, Pandas, Streamlit, Matplotlib, Seaborn

Technical Skills

Programming Languages: Python, SQL, C++, Java, C#

Data Engineering: ETL/ELT Pipelines, Data Modeling, Data Warehousing, Star Schema, Batch Processing, Streaming Processing, Data Validation

Big Data & Distributed Processing: Apache Spark, PySpark, Apache Kafka, Hadoop, Hive, Spark Structured Streaming, Apache Flink, HBase, Elasticsearch, Big Data Analytics

Data Orchestration & Automation: Apache Airflow, dbt, GitHub Actions, CI/CD, Apache ZooKeeper, Apache Flume

Cloud & Azure: Azure Synapse Analytics, Azure Data Factory, Azure Databricks, Dedicated SQL Pools, Azure Stream Analytics

Databases & Data Storage: PostgreSQL, MySQL, SQL Server, SQLite, MongoDB, ClickHouse, Delta Lake, MinIO **Distributed**

Systems & DevOps: Kubernetes (K3s), Docker, Docker Compose, Tailscale, Git, GitHub, Linux

Data Analytics & ML: Pandas, NumPy, Scikit-learn, CatBoost, NLTK, Matplotlib, Seaborn

Development & BI Tools: Jupyter Notebook, JupyterLab, Streamlit, REST APIs, Spark Thrift Server, Pipeline Monitoring

Human Languages: Arabic (Native), English (Professional Working Proficiency)

Education

Bachelor of Computers and Artificial Intelligence

Expected: 2027

Minia National University, Egypt

Relevant Coursework: Data Structures, Data Mining, Algorithms, Database Systems, OOP, Machine Learning

Leadership & Activities

- Mentor at ICPC MNU Team — guided competitive programming students in problem-solving and algorithms.
- Mentor at MENTOR Network — supported junior students in career development and technical growth.
- Admin at Student Activity — managed organizational and academic community events with student committees.